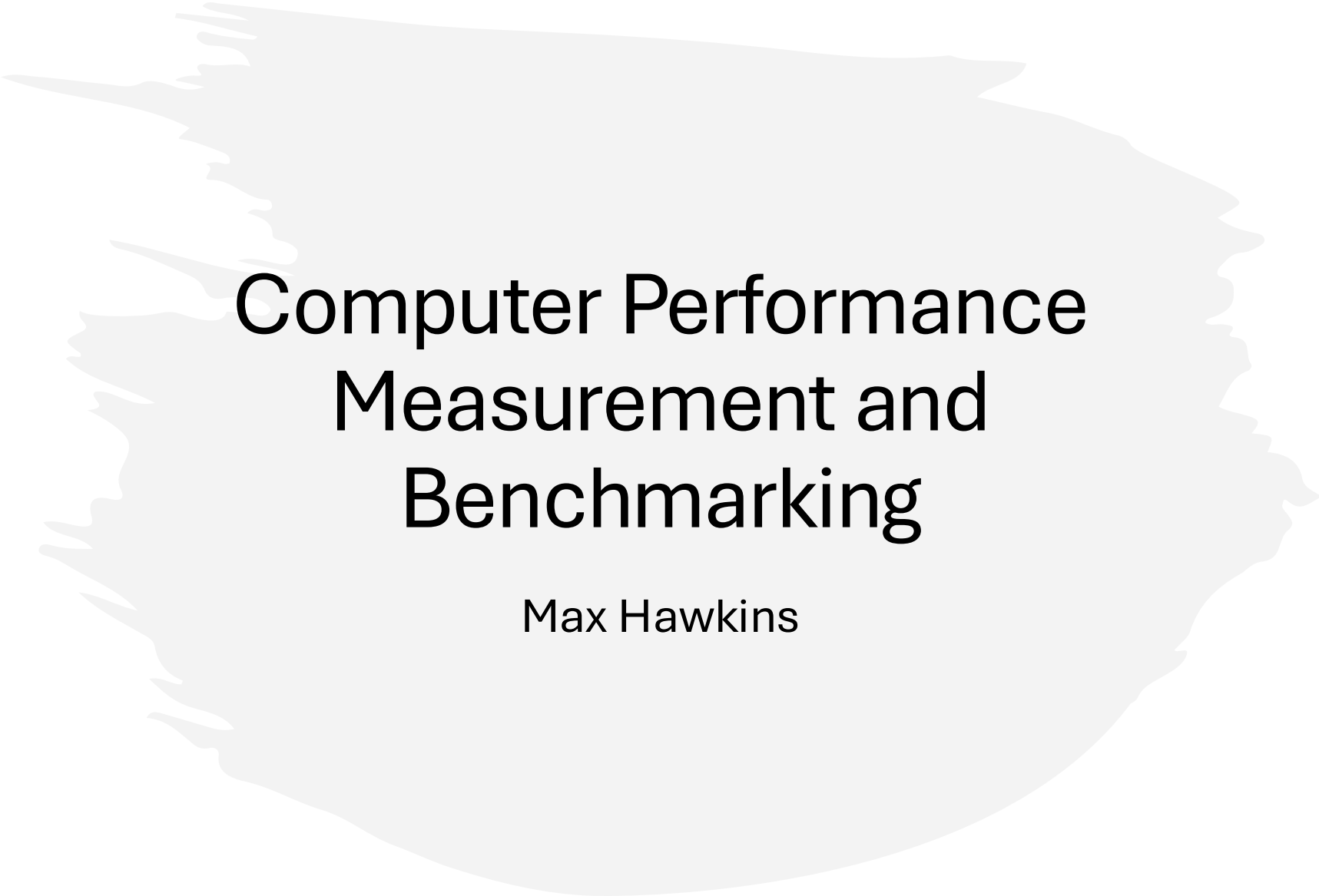




Computer Performance Measurement and Benchmarking

Max Hawkins

What is performance?

Making broad generalizations:

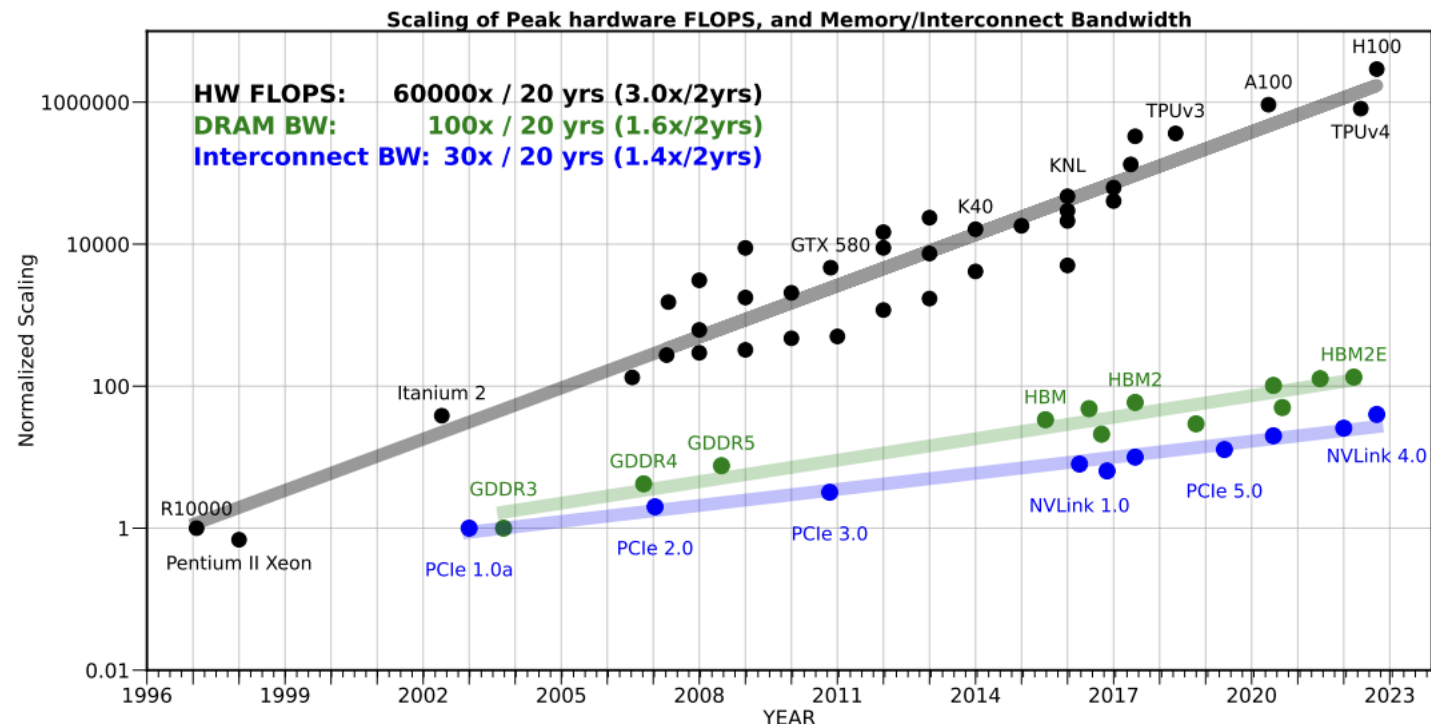
- Hardware architects: Sub-system performance
- Hardware vendors: Application-agnostic system performance
 - Memory bandwidth, operation throughput, etc
 - If selling specialized systems, can focus on applications (e.g. AI)
- Application developers: Application performance
 - Images processed per second, output tokens per second, time to first token

Floating-point Operations per Second (flop/s)

- De-facto standard computing performance metric
- Measures the operation throughput
 - Has integer (iop/s), posit (pop/s), and generic (op/s) version
 - Alternative: Instructions per second
- Unit: a quantity chosen as a standard in terms of which other quantities may be expressed
 - E.g. flop, operation, instruction, meter, second, flop/s, bit/s, etc

Exascale vs Exaflop

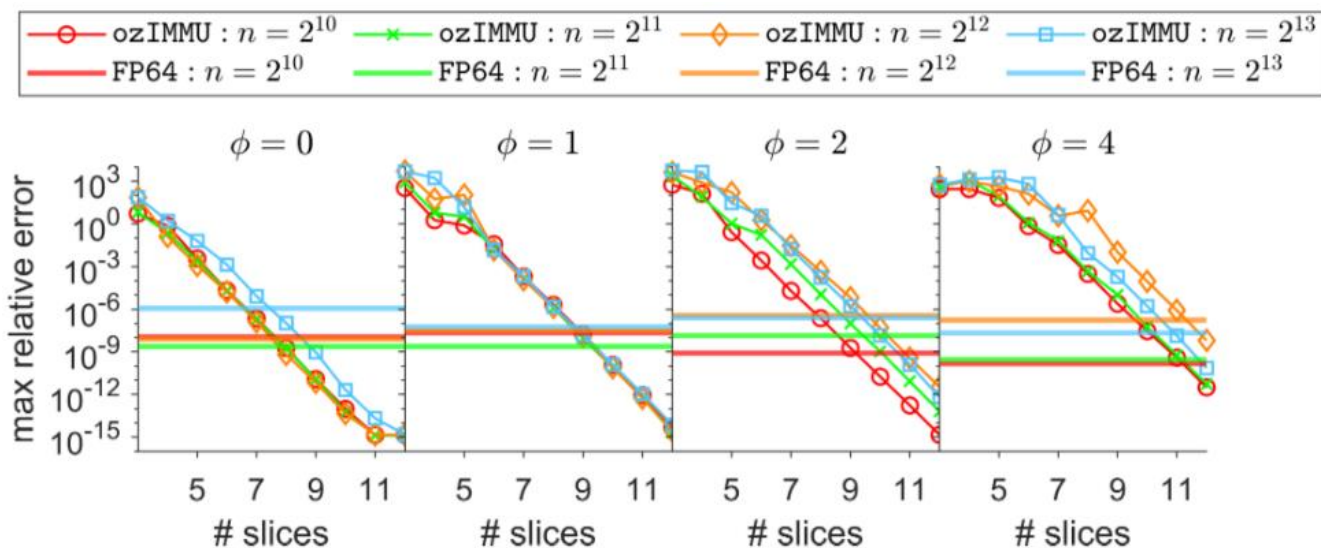
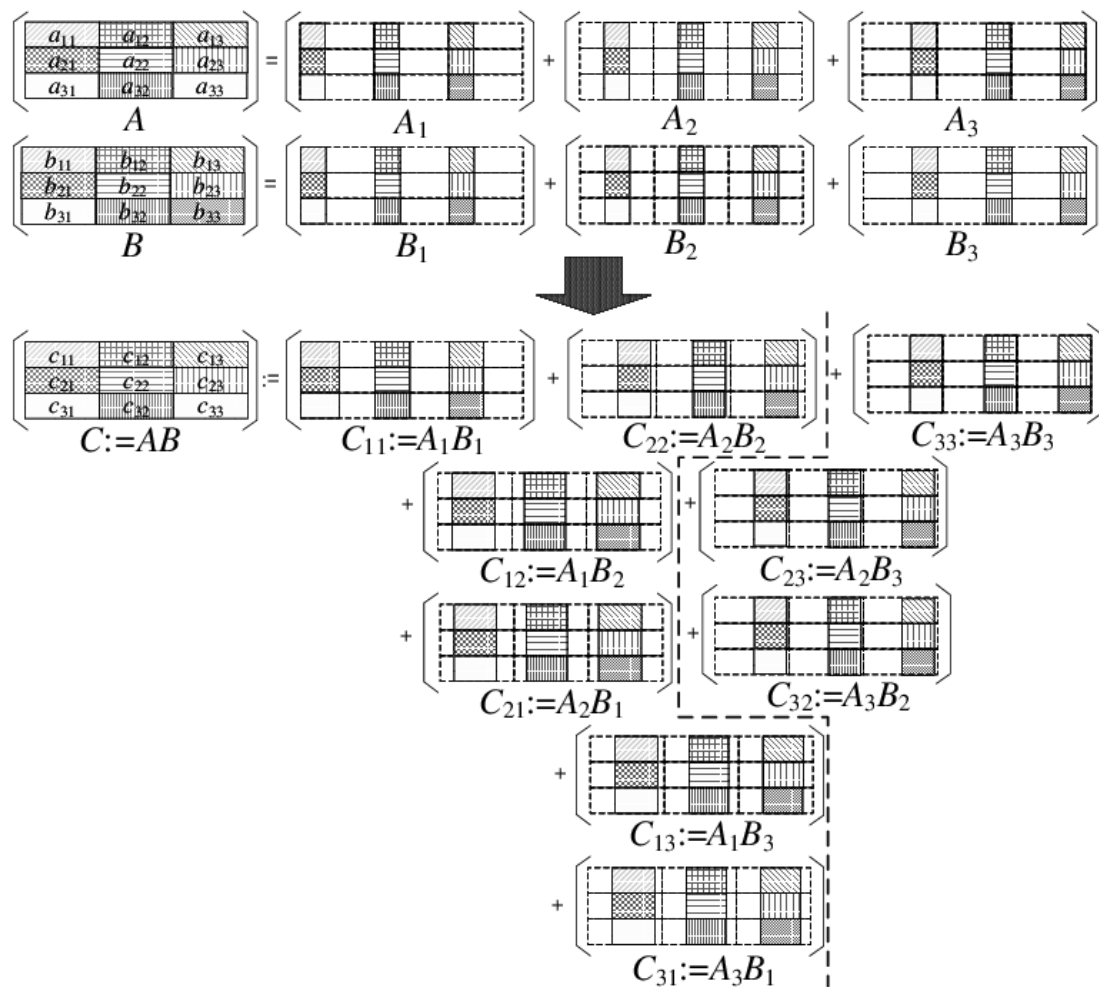
- Frontier is an exaflop computer
 - Can perform $> 1 \times 10^{18}$ flops per second (FP64 with HPL)
- Is Frontier an exascale computer?
 - Exascale = Exaflop?
- Memory?
- Network?



What even is a flop?

- Historically, IEEE-754 FP64 multiplies and adds
 - A narrow but concrete definition...with decreasing prevalence
- Low-precision data types proliferated
 - FP32, 24, 16, 8, 6, 4, 2, 1, ...
 - Quantized AI models
- Operations
 - Just multiplies and adds? Divides?
 - Systolic array (tensor core/TPU) depth effect
- Sparsity
- Redundant encodings
- Emulation
 - Ozaki scheme – performing FP64* arithmetic with Int8 operations

Ozaki Scheme



Benchmarks

- HPL – Dense linear system solver in FP64
 - Favors dense matmuls – tensor cores/TPUs/etc
 - <https://www.netlib.org/benchmark/hpl/>
- HPCG – Sparse iterative solver in FP64
 - <https://github.com/hpcg-benchmark/hpcg>
- MLPerf – Machine-learning inference
 - <https://arxiv.org/pdf/1911.02549>

Lists (and bragging rights)

- Top500
 - Uses HPLinpack
- Green500
 - Uses Top500 results normalized by power draw
- Graph500
 - <https://graph500.org>
 - “Not all workloads are dense matmul” – Someone (probably)

Top500

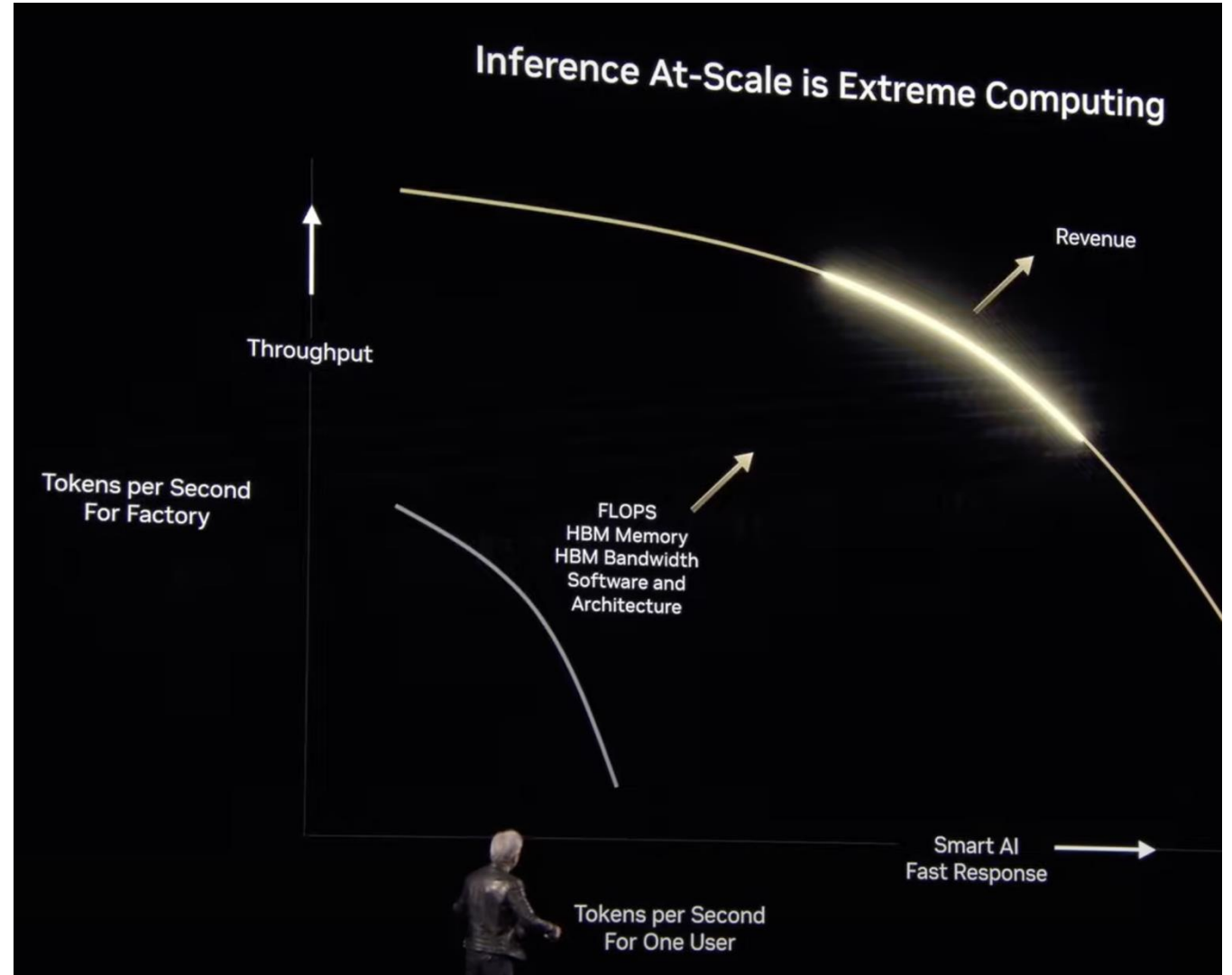
Rank	System	Cores	Rmax (PFlop/s)	Rpeak (PFlop/s)	Power (kW)
1	El Capitan - HPE Cray EX255a, AMD 4th Gen EPYC 24C 1.8GHz, AMD Instinct MI300A, Slingshot-11, TOSS, HPE DOE/NNSA/LLNL United States	11,039,616	1,742.00	2,746.38	29,581
2	Frontier - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE Cray OS, HPE DOE/SC/Oak Ridge National Laboratory United States	9,066,176	1,353.00	2,055.72	24,607
3	Aurora - HPE Cray EX - Intel Exascale Compute Blade, Xeon CPU Max 9470 52C 2.4GHz, Intel Data Center GPU Max, Slingshot-11, Intel DOE/SC/Argonne National Laboratory United States	9,264,128	1,012.00	1,980.01	38,698
4	JUPITER Booster - BullSequana XH3000, GH Superchip 72C 3GHz, NVIDIA GH200 Superchip, Quad-Rail NVIDIA InfiniBand NDR200, RedHat Enterprise Linux, EVIDEN EuroHPC/FZJ Germany	4,801,344	793.40	930.00	13,088
5	Eagle - Microsoft NDv5, Xeon Platinum 8480C 48C 2GHz, NVIDIA H100, NVIDIA Infiniband NDR, Microsoft Azure United States	2,073,600	561.20	846.84	

Green 500

Rank	TOP500 Rank	System	Cores	Rmax (PFlop/s)	Power (kW)	Energy Efficiency (GFlops/watts)
1	259	JEDI - BullSequana XH3000, Grace Hopper Superchip 72C 3GHz, NVIDIA GH200 Superchip, Quad-Rail NVIDIA InfiniBand NDR200, ParTec/EVIDEN EuroHPC/FZJ Germany	19,584	4.50	67	72.733
2	148	ROMEO-2025 - BullSequana XH3000, Grace Hopper Superchip 72C 3GHz, NVIDIA GH200 Superchip, Quad-Rail NVIDIA InfiniBand NDR200, Red Hat Enterprise Linux, EVIDEN ROMEO HPC Center - Champagne-Ardenne France	47,328	9.86	160	70.912
3	484	Adastra 2 - HPE Cray EX255a, AMD 4th Gen EPYC 24C 1.8GHz, AMD Instinct MI300A, Slingshot-11, RHEL, HPE Grand Equipement National de Calcul Intensif - Centre Informatique National de l'Enseignement Suprieur (GENCI-CINES) France	16,128	2.53	37	69.098
4	183	Isambard-AI phase 1 - HPE Cray EX254n, NVIDIA Grace 72C 3.1GHz, NVIDIA GH200 Superchip, Slingshot-11, HPE University of Bristol United Kingdom	34,272	7.42	117	68.835

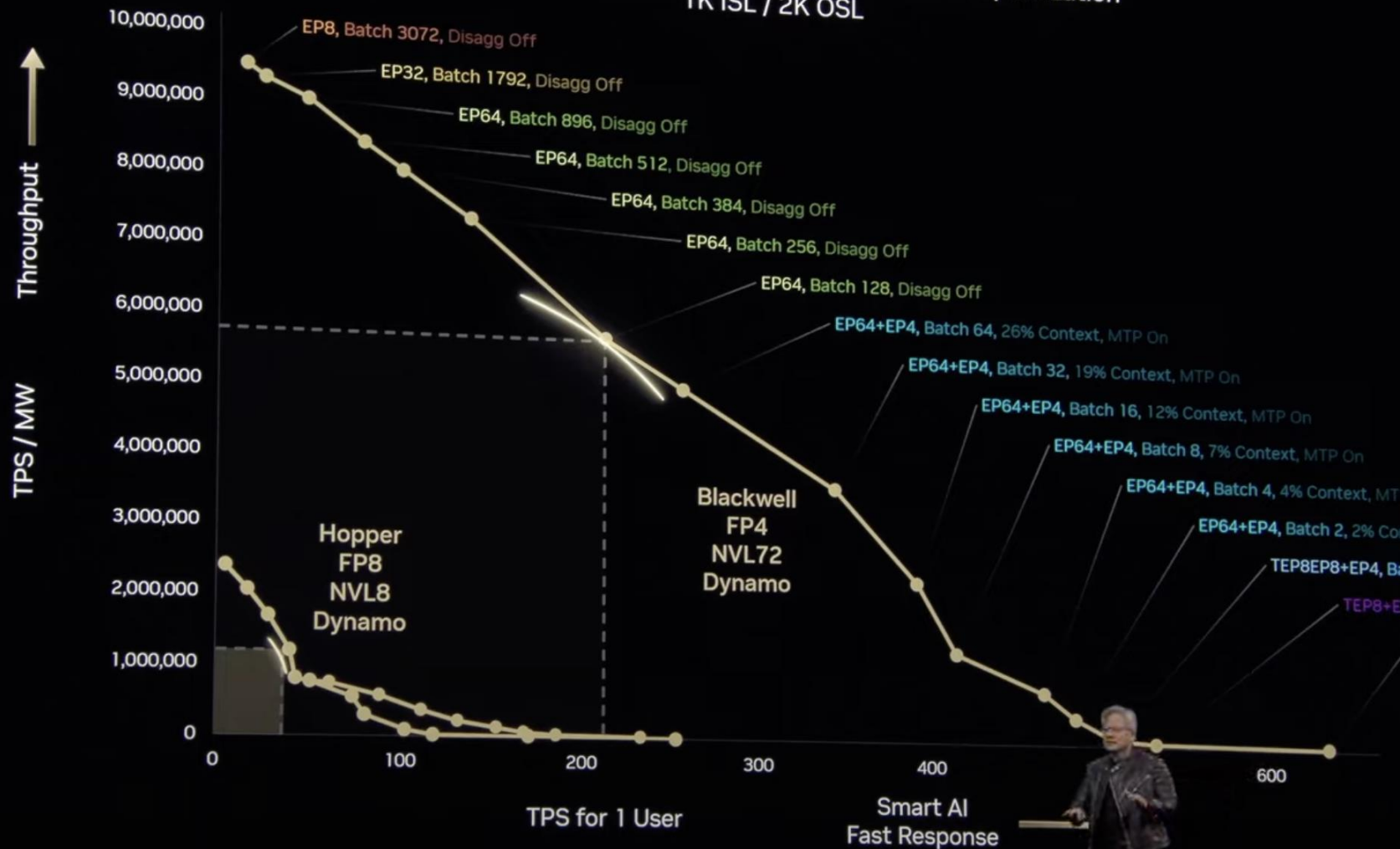
LLM Tokens as Commodities

- Financial value: \$ per token
- Throughput: tokens per second
→ \$ per second
- AI factories: Converting electric power to money
 - “Computational alchemy”



Blackwell 25X Hopper

FP4, NVL72, Dynamo, and TRT-LLM Continuous Optimization
1K ISL / 2K OSL



Aside: Back to bits research project

